

Course Code	ME-41XXN	Course Name	AI&ML Applications in Mechanical Engineering	Course Category	OE-4 (7 th Sem)	L	T	P
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Pre-requisite Courses	1. Mathematics I (MA-1101) 2. Introduction to Computing (CS-1101) 3. Mathematics II (MA-1201) 4. Mathematics III (MA-2101) 5. Numerical Methods in Engineering (ME-2102) or, equivalent subjects	Co-requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department		Mechanical Engineering		Data Book / Codes / Standards	Nil

Course Objective	1) To make the students understand the basic principles of Artificial Intelligence towards problem solving, inference, perception, knowledge representation, and learning. 2) To provide the students foundation in mathematical concepts related to Machine Learning 3) To equip the students to choose appropriate classification technique to perform classification, model building and evaluation 4) To make the students able to model and optimize real-life problems related to mechanical engineering using previously learned concepts
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Module	Syllabus	Duration (class-hour)	Module Outcome
Artificial Intelligence	Introduction, intelligent agent (Environment, Rationality, Good Behaviour, Structure of agent), Solving problems by Search (Formulating problems, Searching for Solutions, Uninformed Search Strategies, Breadth-first search, Depth-first search, Searching with Partial Information, Informed (Heuristic) Search Strategies, Greedy best-first search, A* Search, CSP, Means-End-Analysis), ADVERSARIAL SEARCH (Games, The Mini-Max algorithm, Alpha-Beta Pruning), LOGICAL AGENTS (Knowledge-Based agents, Logic, Propositional Logic, Reasoning Patterns in Propositional Logic, Resolution, Forward and Backward chaining), FIRST ORDER LOGIC (Syntax and Semantics of First-Order Logic, Using First-Order Logic, Knowledge Engineering in First-Order Logic), UNCERTAINTY (Acting under Uncertainty, Basic Probability Notation and Axioms of Probability, Joint Distributions, Independence, Bayes' Rule, Representing Knowledge in an Uncertain Domain),	12	After successful completion of the module, the students will be able to UNDERSTAND the working principles of the intelligent logical agent and their real-life implementation. (CO1)
Learning Methods	Statistical Learning, Learning with Complete Data, Learning with Hidden Variables, Rote Learning, Learning by Taking Advice, Learning in Problem-solving, learning from Examples: Induction, Explanation-based Learning, Discovery, Analogy, Formal Learning Theory, Neural Net Learning and Genetic Learning. Expert Systems: Representing and Using Domain Knowledge, Expert System Shells, Explanation, Knowledge Acquisition, Lazy Lerner	06	After successful completion of the module, the students will be able to UNDERSTAND the architecture and working mechanism of the different models of machine learning. (CO2)
Artificial Neural Network (ANN)	Basic NN Architectures, Learning algorithms and paradigms, Learning, Single layer and multilayer perceptions, back propagation network, SOM, Radial basis Function Networks, Elman-Jordan Recurrent Neural Networks, Convolutional and Deep Neural Network and related algorithms, Applications in Thermal, Design and Manufacturing problems	05	After successful completion of the module, the students will be able to APPLY Artificial Neural Network to real problems in mechanical engineering. (CO3)

Fuzzy Logic System	Fuzzy Set Theory, Fuzzy Relation, Fuzzy Logic and Approximate Reasoning, Fuzzy logic system design, Applications in Thermal, Design and Manufacturing problems	04	After successful completion of the module, the students will be able to APPLY Fuzzy Logic to real problems in mechanical engineering. (CO4)
Optimization	Introduction, Single, Multi and Many Objective Optimization, Pareto Optimal Front, Non-linear Unconstrained and Constrained Multivariable Optimization, Multi-Variable Constrained Numerical Algorithms, Nature Inspired Optimization Techniques: Genetic algorithms (GAs), GA operators: Crossover, Mutation etc., Swarm Intelligence based techniques (Particle Swarm Optimization, Firefly algorithm, Flower pollination algorithm, Crow search algorithm etc.), Queuing models: General characteristics, Markovian queuing model, Queue discipline.	08	After successful completion of the course, the students will be able to APPLY different optimization techniques on different problems in mechanical engineering. (CO5)
Class Test		01	
Total Hours		36	

Course Outcome	After successful completion of the course, the students will be able to: 1. UNDERSTAND the working principles of the intelligent logical agent and their real-life implementation. 2. UNDERSTAND the architecture and working mechanism of the different models of machine learning. 3. APPLY Artificial Neural Network to real problems in mechanical engineering. 4. APPLY Fuzzy Logic to real problems in mechanical engineering. 5. APPLY different optimization techniques on different problems in mechanical engineering.
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Learning Resources	Text book 1. Neural networks: A comprehensive foundations, Simon Haykin, Pearson Education 2. Soft computing: fundamentals and applications, Dilip K. Pratihari, Alpha Science International Ltd. 3. Artificial Intelligence - A Modern Approach, Stuart Russell and Peter Norvig, Pearson Education. 4. Genetic Algorithms in Search, Optimization and Machine Learning, David E. Goldberg, Addison Wesley Reference book 1. Artificial neural networks, B. Vegnanarayana, Prentice Hall of India 2. Neural networks in Computer intelligence, Li Min Fu, Tata McGraw Hill 3. Neural networks, James A Freeman and David M Skapura, Pearson Education 4. An Introduction to Genetic Algorithms, Melanie Mitchell, MIT Press 5. Uncertain Rule-Based Fuzzy Logic Systems: Introduction and New Directions, Jerry M. Mendel, Prentice Hall 6. Fuzzy Logic with Engineering Applications, Timothy. J. Ross, Wiley 7. Fuzzy Sets and Fuzzy Logic – Theory and Applications, G. J. Klir and Bo Yuan, Prentice Hall India 8. Artificial Intelligence, Elaine Rich, Kevin Knight and Shivashankar B Nair, McGraw Hill 9. A Neuro-Fuzzy Synergism to Intelligent System, C.T. Lin and George Lee, Prentice Hall 10. Artificial Intelligence: A New Synthesis, Nils J Nilsson, Morgan Kaufmann Publications 11. Introduction to Artificial Intelligence & Expert Systems, Dan W Patterson, Prentice Hall of India 12. Artificial Intelligence, S Kaushik, Cengage Learning
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